Aldo Acevedo

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Skills

- **Programming Languages:** Python, Rust, C++, Verilog
- **Libraries:** PyTorch, OpenCL, OpenMP, Huggingface (transformers, diffusers), OpenCV, Pandas, NumPy
- **Technologies:** Git, Linux, WebGPU, WebGL, Arduino

Education

National Taiwan University of Science and Technology
3-semester exchange, GPA: 4.27/4.30 — *2025 — Present*

Universidad Politécnica Taiwán-Paraguay
B.E. in Information Engineering, GPA: 4.56/5.00 — *2023 — Present*

- **Relevant coursework:** Parallel and Distributed Computing, Digital Logic Design with Verilog EDA, Computer Architecture, Operating Systems.

Experience

Research Assistant — Jul 2025 — Present
Artificial Vision Lab, National Taiwan University of Science and Technology — *Taipei, Taiwan*

- Working on skeleton-based action recognition, VLMs for age estimation and defect detection in industrial settings.

Software Engineer — May 2022 — Jun 2024
Freelance — *Remote*

- Assembled chunk-based terrain system with procedural vegetation, water and in-game painting/sculpting tools.
- Created pre-rendering system eliminating real-time skeletal animation, allowing low-end Chromebooks to run web game.
- Built Kindle highlights parser with EPUB export, for book summary generation in Rust/WASM with Svelte frontend.
- Implemented GLTF scene viewer and forward-rendering graphics engine in WebGL and Rust/WASM.

Software Engineer — 2021 – 2022 (summer jobs)
Posibillian Tech S.A. — *Asunción, Paraguay*

- Led development of platformer mobile game shipped to 10,000+ players. Implemented mechanics, procedural animation.
- Designed API and devtools for Cubie World mobile game, reducing app memory usage from **800MB** to **80MB**.

Publications

- Y.H. Chiu, **A. A. Onieva**, G.S. J. Hsu, J.H. Kang. "Skeleton-Based Motion Pattern Recognition for Sarcopenia Risk Assessment." Under review, 2026.
- Y.H. Chiu, **A. A. Onieva**, G.S. J. Hsu. "AgeMoVLE: Age Estimation with Mixture of VL Experts." Under review, 2026.

Projects

- **`parrl`**: Implemented REINFORCE policy gradient in C++ with *no* ML libraries. With vectorized environments and parallel backprop, achieved **212x** speedup over serial PyTorch version, converging in **2.3 seconds** *instead* of **8 minutes**.
- **MonkeyOCR-Qwen3**: QLoRA fine-tuned Qwen3-VL-Instruct-2B to replace MonkeyOCR backbone, reducing parameters by **1.5x**, maintaining 96% of original performance. Developed batching mechanism, maximizing GPU utilization and reducing OmniDocBench inference time from **2 hours** to **50 minutes** on an RTX 3090.
- **FastComposer-mini:** Redesigned FastComposer, a Stable Diffusion-based image generation model. Replaced backbone with `tiny-sd`, and achieved a **1.5x inference speedup**, **11x size reduction**, while maintaining comparable quality.
- **Implementing Feature Visualization**: Wrote explainer article showcasing learned features of ResNet-50, with step-by-step replication in PyTorch.
- **`souvlaki` (156,000+ downloads**, 121 stars): Author and maintainer of Rust library providing a unified abstraction for OS media controls, covering MPRIS/D-Bus, Windows SMTC, and macOS MediaPlayer API. Used by **407+** dependents.